

Pleiotropy

This exercise was designed to give practical experience identifying cross phenotype associations using both univariate and multivariate methods and then dissecting these cross phenotype associations to determine if they show evidence of biological and/or mediated pleiotropy.

A population-based dataset with 3000 subjects and two quantitative traits (Trait 1 and Trait 2) along with 2000 SNPs on one chromosome were simulated. Let's assume that Trait 1 was measured 20 years prior to Trait 2 (i.e. Trait 1 will act as the mediator in our mediation analysis). The two quantitative traits are correlated and there are markers associated with one or both phenotypes as well as unassociated.

Adapted from an exercise by Andrew DeWan, PhD, MPH.

Data overview

The data is in PLINK binary-file format under `data/ : pleiotropy_exercise.bed , pleiotropy_exercise.bim , and pleiotropy_exercise.fam .` `data/pleiotropy_exercise_phenotypes.txt` is a ready-made phenotype file. The dataset has already been QC-ed.

How this notebook is organized

1. **Univariate analysis** — test each SNP against Trait 1 and Trait 2 separately, visualize overlap, and LD-clump the genome-wide significant hits.
2. **Multivariate analysis** — jointly test both traits to find additional suggestive/significant regions the univariate scans might have missed.
3. **Mediation analysis** — for the SNPs identified above, test whether their effect on Trait 2 runs *through* Trait 1 (mediated pleiotropy) or acts independently on both traits (biological pleiotropy).

Summary table

A summary table is included at the end of the exercise that you will want to fill out as you are working through this exercise. This will help keep track of the SNPs you select for the mediation analysis as well as the interpretation of the results at the end of the exercise.

Program documentation

- **PLINK:** [1.90 docs](#)
- **R:** [r-project.org](https://www.r-project.org/) — from within R, `help()` documents any command

References

- Chang CC, Chow CC, Tellier LCAM, Vattikuti S, Purcell SM, Lee JJ (2015) Second-generation PLINK: rising to the challenge of larger and richer datasets. *GigaScience*, 4.
- Purcell S, Neale B, Todd-Brown K, Thomas L, Ferreira MAR, Bender D, Maller J, Sklar P, de Bakker PIW, Daly MJ & Sham PC (2007) PLINK: a toolset for whole-genome association and population-based linkage analysis. *American Journal of Human Genetics*, 81:559-575.
- R Core Team (2025), R: A Language and Environment for Statistical Computing, R Foundation for Statistical Computing
- Tingley D, Yamamoto T, Hirose K, Keele L, Imai K (2014). "mediation: R Package for Causal Mediation Analysis." *Journal of Statistical Software*, 59(5), 1–38.
<https://www.jstatsoft.org/v59/i05/>.

Kernel: run the code cells in this exercise with the **Bash** kernel.

1. Univariate Analysis

Conduct a univariate analysis (using `--linear`) in PLINK for both datasets and both traits

Note: You will need to use the `--pheno/--pheno-name` commands to specify the phenotype file and phenotype name.

Run the analysis using PLINK

```
In [ ]: # Trait 1: --linear runs linear regression per SNP; --sex includes sex as a
plink \
  --bfile ./data/pleiotropy_exercise \
  --pheno ./data/pleiotropy_exercise_phenotypes.txt \
  --pheno-name Trait1 \
  --sex \
  --linear \
  --out ./output/Trait1

# --linear/--sex produce one row per SNP per term (ADD, SEX, ...); keep the
# header plus just the SNP (ADD) rows for the downstream steps
grep 'TEST' ./output/Trait1.assoc.linear > ./output/Trait1_snp.assoc.linear
grep 'ADD' ./output/Trait1.assoc.linear >> ./output/Trait1_snp.assoc.linear
```

```
In [ ]: # Trait 2: same analysis, repeated for the second phenotype
plink \
```

```
--bfile ./data/pleiotropy_exercise \
--pheno ./data/pleiotropy_exercise_phenotypes.txt \
--pheno-name Trait2 \
--sex \
--linear \
--out ./output/Trait2

# Keep the header plus just the SNP (ADD) rows, as above
grep 'TEST' ./output/Trait2.assoc.linear > ./output/Trait2_snp.assoc.linear
grep 'ADD' ./output/Trait2.assoc.linear >> ./output/Trait2_snp.assoc.linear
```

Visualize data and identify overlapping signals

Try visualizing the data by creating a Hudson plot and identify genome-wide significant SNPs ($p < 5 \times 10^{-8}$) that overlap for both traits using R. This will give you some sense of the overlapping signals between the two association analyses.

This will be ran using an external R script

`scripts/generate_plots_overlap_signals_univariate.R`, the code used to generate the plot and identify significant SNPs is provided below:

```
library(hudson)

dat1<-read.table("./output/Trait1_snp.assoc.linear",header=T)
dat2<-read.table("./output/Trait2_snp.assoc.linear",header=T)
names(dat1)<-c("CHR", "SNP", "POS", "A1", "TEST", "NMISS", "BETA",
"STAT", "pvalue")
names(dat2)<-(names(dat1))
gmirror(top=dat1, bottom=dat2, tline=5e-08, bline=5e-08,
        toptitle="Trait1", bottomtitle = "Trait2",
        highlight_p = c(0.00000005,0.00000005), highlighter="green",
        file = './output/pleiotropy_hudson', res = 300, type = 'pdf')

Trait1 <- read.table("./output/Trait1_snp.assoc.linear", header =
T)
Trait2 <- read.table("./output/Trait2_snp.assoc.linear", header =
T)
SigTrait1 <- subset(Trait1, P<0.00000005)
SigTrait2 <- subset(Trait2, P<0.00000005)
intersect(SigTrait1$SNP, SigTrait2$SNP)
```

```
In [ ]: # Generate the Hudson plot (./output/pleiotropy_hudson.pdf) and print any
# genome-wide significant SNPs shared between Trait1 and Trait2 to the console
Rscript ./scripts/generate_plots_overlap_signals_univariate.R
```

LD clumping

As you can see, there are some genome-wide significant SNPs that are adjacent or close to each other. To explore whether or not these are independent associations, let's

perform some simple LD clumping. You will want to carry through the index SNP identified for each clumped region. You will also want to carry through any SNPs from above that were not part of a clumped region.

```
In [ ]: # LD clumping for the genome-wide significant SNPs: --clump-p1/--clump-p2 se
# significance thresholds for index/secondary SNPs, --clump-r2/--clump-kb de
# the LD window, and --clump-replicate/--clump-verbose report whether a clun
# found in one trait's results replicates in the other's
plink \
  --bfile ./data/pleiotropy_exercise \
  --clump ./output/Trait1_snp.assoc.linear,./output/Trait2_snp.assoc.linear \
  --clump-kb 250 \
  --clump-p1 5e-8 \
  --clump-p2 5e-8 \
  --clump-r2 0.2 \
  --clump-replicate \
  --clump-verbose \
  --out ./output/Trait1_Trait2_clump
```

2. Multivariate Analysis

In this section we will use multivariate analysis to see if there are additional SNPs/regions to explore, prior to moving on to dissecting the cross phenotype associations. We will only consider additional regions that are genome-wide suggestive for both phenotypes.

Run the analysis on Traits 1 & 2

We use the `MultiPhen` R package (`mPhen`), which jointly models Trait1 and Trait2 as predictors of each SNP (ordinal regression) and reports a single joint p-value per SNP — the same kind of cross-phenotype signal PLINK's old `--mqfam` multivariate test gave, but from an actively maintained tool.

Note: this tests all 2000 SNPs and takes a couple of minutes to run.

```
In [ ]: Rscript - <<'EOF'
suppressMessages(library(MultiPhen))

# mPhen names genotype columns "chr_pos" by default; remap to the .bim file'
# rs IDs so this output lines up with the SNP naming used everywhere else
bim <- read.table("./data/pleiotropy_exercise.bim",
                  col.names = c("chr", "snp", "cm", "pos", "a1", "a2"))
key <- paste(bim$chr, bim$pos, sep = "_")

geno <- mPhen.readGenotypes("./data/pleiotropy_exercise", opts = mPhen.opti
colnames(geno$genoData) <- bim$snp[match(colnames(geno$genoData), key)]
pheno <- mPhen.readPhenoFiles("./data/pleiotropy_exercise_phenotypes.txt",
                             opts = mPhen.options("pheno.input"))

opts <- mPhen.options(c("regression", "pheno.input"))
```

```

opts$mPhen.scoreTest <- TRUE # score test instead of full likelihood ratio:

# Run in 100-SNP batches: mPhen is designed to be called this way (see
# ?mPhen.readGenotypes), and testing all 2000 SNPs in a single call is
# dramatically slower than the sum of the batched calls.
n <- ncol(geno$genoData)
batch <- 100
chunks <- list()
for (start in seq(1, n, by = batch)) {
  end <- min(start + batch - 1, n)
  r <- mPhen(geno$genoData[, start:end, drop = FALSE], pheno$pheno,
            phenotypes = c("Trait1", "Trait2"), opts = opts)
  chunks[[length(chunks) + 1]] <- data.frame(
    SNP      = dimnames(r$Results)[[2]],
    P        = as.numeric(r$Results["all", , "JointModel", "pvalue"]),
    Beta_Trait1 = as.numeric(r$Results["all", , "Trait1", "beta"]),
    Beta_Trait2 = as.numeric(r$Results["all", , "Trait2", "beta"])
  )
}
out <- do.call(rbind, chunks)

write.table(out, "./output/Trait1_Trait2_multiphen.txt",
            sep = "\t", row.names = FALSE, quote = FALSE)
cat("Wrote", nrow(out), "SNPs to ./output/Trait1_Trait2_multiphen.txt\n")
cat("Genome-wide significant (P<5e-8):", sum(out$P < 5e-8), "\n")
EOF

```

Identify multivariate signals

Identify the intersection of SNPs that are genome-wide significant in the multivariate analysis and at least suggestive for each trait in the univariate analysis, i.e. we want to make sure that both traits are contributing to the multivariate signal.

Select the additional SNPs that are identified from the intersection of the multivariate analysis and genome-wide suggestive lists for both traits that were not in your original list.

`./output/Trait1_Trait2_multiphen.txt` has one row per SNP: `P` is the joint model's p-value (this is the "MV (P)" column in the summary table), and `Beta_Trait1` / `Beta_Trait2` are each trait's contribution to that joint model — use these for the "MV Loading" columns.

```

In [ ]: # Define a function to filter a PLINK results file down to SNPs below a p-value
# threshold: find the "SNP" and "P" columns by name (position varies by file
# type), then print the SNP ID for any row whose P column is numeric and < t
filter_snps() {
  awk -v thresh="$2" '
    NR == 1 {
      for (i = 1; i <= NF; i++) {
        if ($i == "SNP") s = i
        if ($i == "P") p = i
      }
    }
    {
      if ($p < thresh) print $s
    }
  '
}

```

```

    }
    if (!s || !p) { print "missing SNP or P column in " FILENAME > "
    next
    }
    $p + 0 == $p && $p + 0 < thresh { print $s }
    ' "$1" | sort -u
}

# Run the function for each trait: suggestive (p<5e-6) for each univariate t
# genome-wide significant (p<5e-8) for the multivariate test
filter_snps ./output/Trait1_snp.assoc.linear 0.000005 > ./output/sug
filter_snps ./output/Trait2_snp.assoc.linear 0.000005 > ./output/sug
filter_snps ./output/Trait1_Trait2_multiphen.txt 0.00000005 > ./output/sig

# Get the intersection of all three lists (comm requires sorted input, which
# filter_snps already provides via `sort -u`)
comm -12 ./output/sugg_trait1.snps ./output/sugg_trait2.snps \
| comm -12 - ./output/sig_multi.snps \
| tee ./output/shared.snps

```

LD clumping

Re-run the LD clumping with a suggestive threshold to see if these additional SNPs clump with your existing clumps or are new potential regions to explore.

```

In [ ]: # Same clumping as before, but at the looser suggestive threshold (5e-6) so
# newly identified multivariate SNPs can be checked against the existing clu
plink \
--bfile ./data/pleiotropy_exercise \
--clump ./output/Trait1_snp.assoc.linear,./output/Trait2_snp.assoc.linear \
--clump-p1 0.000005 \
--clump-p2 0.000005 \
--clump-r2 0.2 \
--clump-replicate \
--clump-verbose \
--out ./output/Trait1_Trait2_clump_suggestive

```

3. Mediation analysis

Now that you've assembled a candidate list of cross-phenotype SNPs (genome-wide significant in both traits, LD-clump index SNPs, and any additional multivariate hits), this section tests each one for **biological vs. mediated pleiotropy** — see the explanation in the introduction above. For each SNP, we extract its genotype, then fit the two regressions the `mediation` package needs (SNP → Trait1, and Trait1 + SNP → Trait2) and let it decompose the SNP's total effect on Trait2 into the ACME (mediated through Trait1) and ADE (direct, not through Trait1) components.

Extract data for each SNP

For each SNP that you have identified as a cross phenotype association (evidence of overlapping association signals as well as incorporating results from LD clumping and multivariate association) you will need to extract this data from the original plink files and create a genotype file that is coded as 0|1|2 for the genotypes. This can be done in PLINK using the `--recodeA` command and the `--extract` command by providing a file with the list of SNPs. This will give you a `.raw` genotype file with only the SNPs that you will be using in the mediation analysis.

Build that SNP list yourself first — `--extract` needs a plain text file with one SNP ID per line, *not* a results file. List the SNPs you selected above: the genome-wide significant overlaps from Step 1, each LD-clump index SNP, and any additional SNPs from the multivariate step (`./output/shared.snps`). The cell below creates a starter file with the worked example this notebook's mediation step uses (`rs125`) — replace its contents with your own selected SNPs before moving on.

```
In [ ]: # Create the SNP list --extract needs below: one SNP ID per line, no header.
# Replace rs125 with your own selected SNPs (one per line) before running the
# mediation analysis on your actual results.
cat > ./output/mediation_snps.txt <<'EOF'
rs125
EOF
```

```
In [ ]: # --recodeA writes an additive (0/1/2 copies of the minor allele) .raw genotype
# file; --extract restricts it to just the SNPs listed in mediation_snps.txt
plink \
  --bfile ./data/pleiotropy_exercise \
  --recodeA \
  --extract ./output/mediation_snps.txt \
  --out ./output/snps_for_mediation
```

Run mediation analysis

Conduct a mediation analysis in R using the mediation R library. A sample code for 1000 simulations is provided below. In real life application, you will need to replace `SNP` with the variable name for the SNP you are investigating and repeat the code for each SNP that you have selected.

Sample code:

```
library(mediation)
genotypes <- read.table("./output/snps_for_mediation.raw",
header=T)
phenotypes <-
read.table("./data/pleiotropy_exercise_phenotypes.txt", header=T)
combined <- merge(genotypes,phenotypes)
med.fit <- lm(Trait1~SNP, data=combined)
out.fit <- lm(Trait2~Trait1+SNP, data=combined)
med.out <- mediate(med.fit,out.fit,treat="SNP", mediator="Trait1",
```

```
boot=TRUE, boot.ci.type="bca", sims=1000)
summary(med.out)
```

Note: The more simulations (sims) you specify in the med.out step the more accurate the CI and p-value estimates will be, however, this can also be time-consuming. If this step is taking a substantial amount of time (>20 minutes) you may want to reduce the number of simulations for the purposes of completing the exercise.

For this minimal working example we run 50 simulations. Although `mediation` is an R package, don't switch kernels — R is invoked with `Rscript` so the notebook stays on the bash kernel throughout.

```
In [ ]: Rscript - <<'EOF'
library(mediation)

genotypes <- read.table("./output/snps_for_mediation.raw", header=T)
phenotypes <- read.table("./data/pleiotropy_exercise_phenotypes.txt", header=
combined <- merge(genotypes, phenotypes)

# Step 1: does the SNP predict the mediator, Trait1?
med.fit <- lm(Trait1 ~ rs125_0, data=combined)
# Step 2: does Trait1 (plus the SNP) predict the outcome, Trait2?
out.fit <- lm(Trait2 ~ Trait1 + rs125_0, data=combined)

set.seed(42)
# Bootstrap the indirect (ACME, via Trait1) and direct (ADE) effects of the
med.out <- mediate(med.fit, out.fit, treat="rs125_0", mediator="Trait1",
boot=TRUE, boot.ci.type="bca", sims=50)
summary(med.out)
EOF
```

Questions

1. Which of the SNPs have genome-wide significant ($p < 5 \times 10^{-8}$) associations for both traits?
2. Did the multivariate analyses result in additional SNPs that had genome-wide significant cross phenotype associations? Which SNP(s)?
3. For each SNP analyzed in the mediation analysis, determine if there is a significant direct effect which is indicative of some level of biological pleiotropy. Do any of the SNPs exhibit complete mediation?
4. Why do some of the SNPs have negative values for the proportion mediated?

Hint: compare the signs of the ACME and ADE estimates in `summary(med.out)` for those SNPs.

Summary Table

SNP	Beta (Trait 1)	P (Trait 1)	Beta (Trait 2)	P (Trait 2)	MV (P)	MV Loading (Trait 1)	M Loa (T ;